

Advancing Human Motion Tracking via a Graph-Enhanced Multimodal Learning Model

Abstract—Accurate human motion tracking is essential to promote musculoskeletal health, whereas it remains challenging in field studies, especially under harsh, dynamic environment. We propose a graph-enhanced multimodal deep learning model (GNN-MTNet) that fuses wearable IMU sensor data with smartphone video inputs to predict full-body joint angles during occupational tasks. Our model employs 70 bidirectional encoders in dual streams (Bi-LSTM and Bi-GRU), processes both modalities through temporal attention modules (TAM), and integrates anatomical relationships via GATv2 graph neural networks. Unlike traditional autoregressive approaches requiring long prediction history windows, our simplified architecture uses lightweight single-step feedback, significantly reducing error accumulation while maintaining temporal coherence. Experiments on 10 subjects performing representative wind turbine maintenance tasks demonstrate that our GNN-MTNet achieves 37.3% RMSE reduction compared to baselines and maintains robust performance even with 30% input masking. The model's graph-based fusion mechanism enables effective information propagation from available sensors, addressing the critical challenge of biomechanical studies in field environments.

I. INTRODUCTION

Accurate biomechanical motion capture in the field is crucial for studying human movement in real-world, dynamic conditions. Traditional motion capture techniques, such as marker-based systems, provide high reliability in controlled laboratory settings but are not applicable for field use. These systems require specialized cameras, reflective markers, and stable environmental conditions that cannot be met under harsh, highly dynamic environment. A reliable, precise method is thus needed to capture motion despite external disturbances, which is essential for fields like ergonomics, sports science, and rehabilitation engineering.

Markerless video-based approaches, as an alternative, are introduced and frequently used in the field environment. It utilizes computer vision models to detect full-body joint positions without markers to estimate human motion[1]. However, this technology is also facing challenges outside the lab; as they require a clear view of the user, a single camera can easily lose track of a user's limbs due to an occlusion[2]. This makes purely vision-based pose estimation unreliable in the real-world environment. Wearable inertial measurement unit (IMU) technology has also been widely used in field

environments due to their portability and independence from visual conditions [3]. By measuring linear acceleration and angular velocity [4-6], IMU-based systems can provide continuous motion estimates. Nevertheless, relying solely on IMU measurements could result in accumulated drift that is caused by biases and noise in the gyroscope and accelerometer during numerical integration [6], leading to degraded estimation of orientation, velocity and position. Therefore, IMUs are better suited as a complementary sensing modality rather than a stand-alone solution.

In this paper, we propose a multimodal machine learning model that integrates wearable IMU data and smartphone video to predict full-body human motion. IMU and video modalities offer complementary strengths: camera-based pose observations provide global spatial context and corrections for IMU drift, while IMUs can then apply continuous orientation data to reduce visual occlusion. By integrating both modalities, a multimodal model can improve accuracy compared with single-modality methods while maintaining comprehensive joint coverage under challenging field conditions. Prior studies have shown evidence of such synergy, e.g., combining IMUs with phone cameras yielded more accurate estimates of knee joint loading than using either modality alone[7]. Here, we extend this multimodal approach to full body pose estimation for dynamic complex industrial tasks. Our proposed model creates a unified representation of human motion by jointly learning from IMU signals and video-derived pose features, which enables more comprehensive joints tracking and the capture of more complex movements than existing wearable-only or video-only systems. Specifically, our contributions are threefold:

1. We propose a graph-enhanced dual-encoder architecture that leverages multiple bidirectional recurrent neural networks to process IMU signals and keypoint data extracted from dual-view video, thereby capturing fine-grained temporal dynamics across modalities.
2. We introduce a multimodal anatomical graph and employ GATv2-based message passing to explicitly model anatomical relationships across heterogeneous sensing modalities, establishing a principled graph-based framework for cross-modal information propagation and structured reasoning.
3. We design a lightweight autoregressive mechanism with

single-step feedback that eliminates long prediction history windows, dramatically reducing exposure bias

II. METHOD

A. Experiment Setup

1) *Data collection Protocol*: 10 subjects were recruited, and the protocol was approved by the Institutional Review Board of Texas A&M University. The subjects were asked to perform four repetitions of three representative occupational tasks, with task1: symmetric lifting, task2: climbing three rungs, and task3: ladder stepping[3]. Data was captured using 11 flexible IMU sensors for linear acceleration and angular velocity (sampling rate: 62.5 Hz), 40 reflective markers for ground truth motion tracking, and two iPhones positioned at 30-45 degree angles to record dual-view videos. The experimental setup is illustrated in Fig. 1. Specific static and synchronization trials were conducted to calibrate the IMU sensors before data collection.

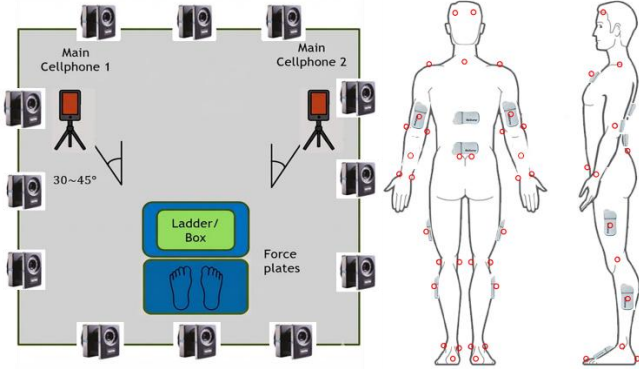


Fig. 1. Experiment setup showing dual-camera configuration, and representative occupational tasks performed by subjects.

2) *Data preprocessing*: the preprocessing pipeline consists of four main stages:

a) *Filtering*: IMU data were filtered at 10 Hz using a zero-lag fourth-order Butterworth low-pass filter to reduce high-frequency noise while preserving motion dynamics. Marker data was filtered at 5 Hz using the same filter type for ground truth reference. OpenPose data were filtered at 15 Hz using a zero-lag fourth-order Butterworth filter after keypoint extraction.

b) *Keypoint Extraction*: OpenPose (version 1.7.0, Body_25 model)[8], a joint center detection algorithm, was used to estimate the 2-D position of 13 keypoints (elbow, shoulder, hip, mid-hip, knee, ankle, heel) from each frame of the dual-view videos. Each keypoint is represented by (x, y, c) coordinates where c is the confidence score.

c) *Coordinate Transformation*: the mid-hip point was defined as the origin to remove global translation effects. This ensures that 2-D joint trajectories represent motion relative to the subject's body rather than position within the camera frame, providing translation-invariant features.

d) *Synchronization*: cross-correlation results were used to synchronize the marker-based system and video-based system.

B. Proposed Deep learning model

In this section, we present our Graph-enhanced Multi-modal Temporal Network (GNN-MTNet) for human joint angle prediction. The model jointly learns from both IMU sensor data and video-based keypoint data using a dual-encoder architecture integrated with graph neural networks. Unlike traditional autoregressive approaches with long prediction history windows, our model employs a lightweight autoregressive mechanism with single-step feedback, significantly reducing error accumulation while preserving temporal coherence. The overall architecture is illustrated in Fig. 2.

1) *Multi-modal Dual Encoder System*: Our model processes two complementary modalities: IMU sensor signals from 11 body-mounted sensors, and video keypoint data extracted from two cameras using OpenPose.

Input Modalities: Each IMU sensor generates two types of signals: 3-axis accelerometer and 3-axis gyroscope data. For video data, OpenPose extracts 13 keypoints (l_elbow, r_elbow, l_shoulder, r_shoulder, l_hip, r_hip, mid_hip, l_knee, r_knee, l_ankle, r_ankle, l_heel, r_heel) from two camera views. Each keypoint is represented by (x, y, c) coordinates where c is the confidence score, yielding $13 \times 2 \times 3 = 78$ video features. In total, the model processes 35 input channels (11 accelerometer channels, 11 gyroscope channels, and 13 video keypoint channels) with 144-dimensional features (66 from IMU + 78 from video).

Dual Encoder Architecture: To fully exploit the temporal dynamics, we employ 70 bidirectional recurrent encoders organized in two parallel streams, including 35 Bi-LSTM encoders (11 for IMU-accel, 11 for IMU-gyro, 13 for video) and 35 Bi-GRU encoders with identical input configuration. For each input channel i , the encoder processes a window of $L = 64$ timesteps:

$$\mathbf{h}_i^{\text{LSTM}}, \mathbf{c}_i^{\text{LSTM}} = \text{Bi-LSTM}_i(\mathbf{x}_i^{1:L})$$

$$\mathbf{h}_i^{\text{GRU}} = \text{Bi-GRU}_i(\mathbf{x}_i^{1:L})$$

where $\mathbf{x}_i^{1:L}$ represents the input sequence for channel i . Each encoder outputs hidden states $\mathbf{h}_i \in \mathbb{R}^{L \times 128}$ (128 is the hidden dimension after averaging bidirectional outputs).

2) *Temporal Attention Module (TAM)*: each encoder output is processed by a dedicated TAM to extract the most relevant temporal features. For encoder i , the TAM computes attention weights over the temporal dimension:

$$\mathbf{e}_{i,t} = \mathbf{v}_a^T \tanh(\mathbf{W}_a \mathbf{h}_{i,t} + \mathbf{b}_a)$$

$$\alpha_{i,t} = \frac{\exp(\mathbf{e}_{i,t})}{\sum_{t'=1}^L \exp(\mathbf{e}_{i,t'})}$$

$$\mathbf{z}_i = \sum_{t=1}^L \alpha_{i,t} \mathbf{h}_{i,t}$$

where $\mathbf{h}_{i,t}$ is the encoder output at timestep t , $\mathbf{W}_a, \mathbf{v}_a, \mathbf{b}_a$ are learnable parameters, and $\mathbf{z}_i \in \mathbb{R}^{128}$ is the context vector used for GNN node features.

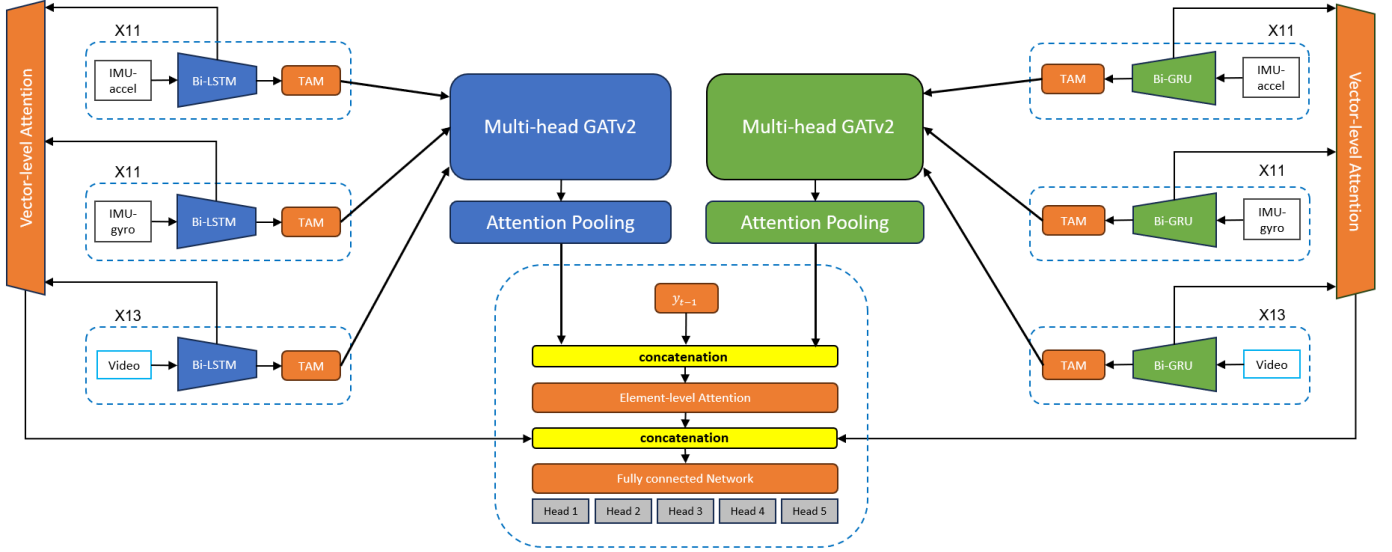


Fig. 2: Overall architecture of GNN-MTNet (Simplified).

3) *Vector-Level Attention (VLA) for Multi-modal Aggregation*: unlike traditional approaches that use VLA for decoder initialization, our simplified architecture employs VLA to directly aggregate encoder outputs for final prediction. VLA operates separately on each modality (accelerometer, gyroscope, and video keypoints) within each encoder path:

For LSTM path:

$$\gamma_i^{\text{accel}} = \frac{\exp(\mathbf{w}_v^T \mathbf{h}_i^{\text{accel,LSTM}})}{\sum_{k=1}^{11} \exp(\mathbf{w}_v^T \mathbf{h}_k^{\text{accel,LSTM}})}$$

$$\mathbf{v}^{\text{accel,LSTM}} = \sum_{i=1}^{11} \gamma_i^{\text{accel}} \mathbf{h}_i^{\text{accel,LSTM}}$$

Similarly for gyroscope and pose modalities. The LSTM path produces:

$$\mathbf{v}^{\text{LSTM}} = [\mathbf{v}^{\text{accel,LSTM}} \parallel \mathbf{v}^{\text{gyro,LSTM}} \parallel \mathbf{v}^{\text{pose,LSTM}}] \in \mathbb{R}^{384}$$

For GRU path: An identical VLA process produces $\mathbf{v}^{\text{GRU}} \in \mathbb{R}^{384}$. These aggregated features capture modality-specific patterns and serve as direct inputs to the final predictor, eliminating the need for separate decoder networks.

4) *Graph Neural Network Integration*: The TAM outputs are used to initialize a 40-node graph representing the anatomical structure (Fig. 3). The graph consists of 16 target joint angle nodes (prediction outputs), 11 IMU sensor nodes (triangular in Fig. 3), and 13 video keypoint nodes (rectangular in Fig. 3).

The adjacency matrix $\mathbf{A} \in \mathbb{R}^{40 \times 40}$ is constructed based on anatomical proximity and sensor placement. Node features are constructed as follows. *IMU nodes*: Concatenation of accelerometer and gyroscope TAM outputs: $[\mathbf{z}_i^{\text{accel}} \parallel \mathbf{z}_i^{\text{gyro}}] \in \mathbb{R}^{256}$. *Pose nodes*: TAM output projected to 256D: $\mathbf{W}_p \mathbf{z}_i^{\text{pose}} \in \mathbb{R}^{256}$. *Target nodes*: Learned embeddings or copied from anatomically related source nodes.

We employ a 3-layer GATv2 network[9] with multi-head attention ($H = 4$ heads):

$$\mathbf{h}_j^{(l+1)} = \parallel_{h=1}^H \sigma \left(\sum_{i \in \mathcal{N}(j)} \alpha_{ij}^{(l,h)} \mathbf{w}^{(l,h)} \mathbf{h}_i^{(l)} \right)$$

$$\alpha_{ij}^{(l,h)} = \frac{\exp(e_{ij}^{(l,h)})}{\sum_{k \in \mathcal{N}(j)} \exp(e_{ik}^{(l,h)})}$$

$$e_{ij}^{(l,h)} = \text{LeakyReLU}(\mathbf{a}^{(l,h)T} \text{LeakyReLU}(\mathbf{W}^{(l,h)} \mathbf{h}_i + \mathbf{W}^{(l,h)} \mathbf{h}_j))$$

where $\mathbf{h}_j^{(l)}$ is the node representation at layer l , $\mathcal{N}(j)$ are neighbors of node j , $\parallel$ denotes concatenation across heads, and $\mathbf{W}^{(l,h)}$, $\mathbf{a}^{(l,h)}$ are learnable parameters per head.

Two parallel GATv2 networks process the LSTM and GRU streams separately, preventing premature fusion and preserving complementary temporal representations.

5) *Attention Pooling*: After the final GNN layer, an attention pooling mechanism aggregates all node features from each path:

$$\beta_j^{\text{LSTM}} = \frac{\exp(\mathbf{w}_p^T \mathbf{h}_j^{\text{LSTM,(3)}})}{\sum_{k=1}^{40} \exp(\mathbf{w}_p^T \mathbf{h}_k^{\text{LSTM,(3)}})}$$

$$\mathbf{g}^{\text{LSTM}} = \sum_{j=1}^{40} \beta_j^{\text{LSTM}} \mathbf{h}_j^{\text{LSTM,(3)}} \in \mathbb{R}^{256}$$

Similarly, for GRU path: $\mathbf{g}^{\text{GRU}} \in \mathbb{R}^{256}$.

6) *Element-level Attention (ELA) and Direct Prediction*: the GNN pooling outputs from both paths are concatenated and refined using ELA:

$$\mathbf{f}_{\text{gnn}} = [\mathbf{g}^{\text{LSTM}} \parallel \mathbf{g}^{\text{GRU}}] \in \mathbb{R}^{512}$$

$$\mathbf{w}_{\text{ELA}} = \sigma(\mathbf{W}_e \mathbf{f}_{\text{gnn}} + \mathbf{b}_e)$$

$$\mathbf{f}_{\text{refined}} = \mathbf{w}_{\text{ELA}} \odot \mathbf{f}_{\text{gnn}}$$

where $\odot$ denotes element-wise multiplication and σ is the sigmoid function. The refined GNN features are concatenated

with VLA outputs and the previous prediction for direct prediction:

$$\mathbf{f}_{\text{final}} = [\mathbf{f}_{\text{refined}} \parallel \mathbf{v}^{\text{LSTM}} \parallel \mathbf{v}^{\text{GRU}} \parallel \mathbf{y}_{t-1}] \in \mathbb{R}^{1296}$$

where $\mathbf{y}_{t-1} \in \mathbb{R}^{16}$ is the previous single-step prediction ($\mathbf{k}_{\text{prev}}$), providing lightweight autoregressive feedback without long history windows.

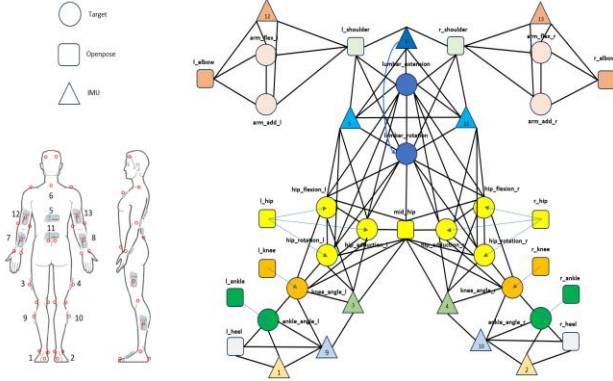


Fig. 3. Graph structure used in GATv2 networks. Directed edges indicate information flow from sensor/keypoint nodes to joint nodes based on anatomical relationships. The graph topology is designed according to human body structure and IMU sensor placement (bottom-left).

7) *Multi-Head Prediction*: the final feature vector is passed through five anatomical-specific heads:

$$\begin{aligned} \mathbf{y}_{\text{hip}} &= \text{FC}_{\text{hip}}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^6 \\ \mathbf{y}_{\text{arm}} &= \text{FC}_{\text{arm}}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^4 \\ \mathbf{y}_{\text{knee}} &= \text{FC}_{\text{knee}}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^2 \\ \mathbf{y}_{\text{lumbar}} &= \text{FC}_{\text{lumbar}}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^2 \\ \mathbf{y}_{\text{ankle}} &= \text{FC}_{\text{ankle}}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^2 \\ \hat{\mathbf{y}}_t &= [\mathbf{y}_{\text{hip}} \parallel \mathbf{y}_{\text{arm}} \parallel \mathbf{y}_{\text{knee}} \parallel \mathbf{y}_{\text{lumbar}} \parallel \mathbf{y}_{\text{ankle}}] \in \mathbb{R}^{16} \end{aligned}$$

The anatomical groups include Hip (6 DoFs): hip flexion/adduction/rotation (hip_fle_r/l , hip_add_r/l , hip_rot_r/l), Arm (4 DoFs): arm flexion/adduction (arm_fle_r/l , arm_add_r/l), Knee (2 DoFs): knee angle (knee_ang_r/l), Lumbar (2 DoFs): lumbar extension/rotation (lumbar_ext and lumbar_rot), and Ankle (2 DoFs): ankle angle (ankle_ang_r/l).

C. Training Strategy

Autoregressive Training: Our architecture uses single-stage training with lightweight autoregressive feedback. The model uses its own previous prediction $\hat{\mathbf{y}}_{t-1}$ as input during both training and inference:

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N \|\hat{\mathbf{y}}_n(\hat{\mathbf{y}}_{n-1}) - \mathbf{y}_n^{\text{GT}}\|^2$$

where N is the number of samples in a batch.

Initial Prediction Strategy: For the first prediction at timestep $t = L$ where no previous prediction exists, we use a statistical

prior $\bar{\mathbf{y}}^{\text{init}} \in \mathbb{R}^{16}$, computed as the mean of all training trials at position $L - 1$:

$$\bar{\mathbf{y}}^{\text{init}} = \frac{1}{M} \sum_{m=1}^M \mathbf{y}_{m,L-1}$$

where M is the number of trials in the training set. This provides a stable starting point consistent across training and inference.

III. RESULTS

A. Dataset and Implementation Details

1) *Dataset Description*: Our experiments were conducted on the dataset described in Section 2, comprising 10 subjects performing three representative occupational tasks, symmetric lifting (Task 1), climbing three rungs (Task 2) and ladder stepping (Task 3). The dataset includes synchronized IMU data (11 body-mounted sensors down sampled to 50Hz) and dual-view video recordings. In total, the dataset contains 235,233 timesteps across multiple trials per subject.

The dataset is split at the subject level, with seven subjects used for training (168,033 samples), one subject reserved for validation (21,224 samples), and two subjects held out for testing (45,976 samples). This subject-level split ensures that the model is evaluated on completely unseen individuals, providing a rigorous test of generalization capability.

2) *Implementation Details*: all models were implemented in PyTorch and trained on NVIDIA A100 80GB GPUs. The training hyperparameters are: Encoder hidden dimension: 128; GNN hidden dimension: 256; Window length L : 64 timesteps; Batch size: 64; Learning rate: Initial learning rate: 1×10^{-3} ; reduced using a ReduceLROnPlateau scheduler; Optimizer: Adam with gradient clipping (max norm: 1.0); Early stopping patience: 20 epochs; Loss function: Mean Squared Error (MSE).

B. Comparison with Baseline Methods

We compared GNN-MTNet against three baseline methods: OpenCap[10], a markerless motion capture system solely based on smartphone videos; MM-LSTM [11], a multi-modal LSTM that performs data fusion through simple concatenation; and TFN-Biomech [7], a Tensor Fusion Network adapted for biomechanical prediction.

1) *Overall Performance*: Fig. 4 presents the comparison across three metrics: RMSE ($^\circ$), relative RMSE (rRMSE), and Pearson correlation coefficient (r). Our GNN-MTNet consistently outperforms all baselines across 16 joint angles. GNN-MTNet achieves 37.3% RMSE reduction compared to MM-LSTM, 4.3% improvement over TFN-Biomech, and 47.4% improvement over OpenCap baseline, with particularly strong performance on arm joints ($r > 0.94$).

2) *Task-Specific Performance*: To evaluate generalization across different occupational tasks, we analyzed performance across three task categories. As shown in Table 1, our model outperforms the baseline methods across all tasks.

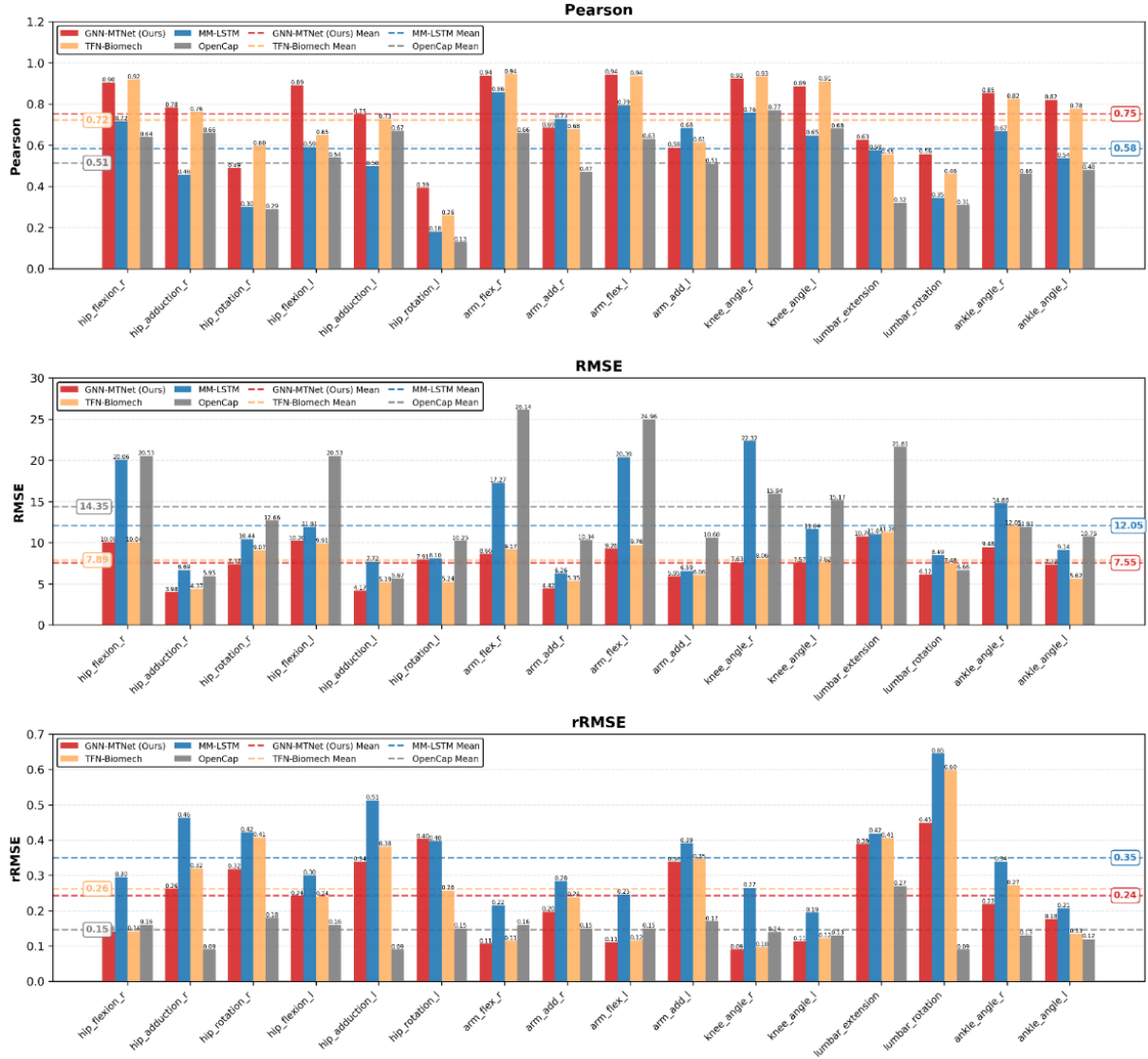


Fig. 4: Overall performance comparison across 16 joint angles. Dashed lines indicate model means.

3) *Temporal Prediction Quality*: Fig. 5 visualizes the temporal prediction quality on representative trials of Task 1. The time-series plots demonstrate that GNN-MTNet (red dashed) closely follows the ground truth (green solid) across all 16 joints, with notably better tracking of rapid movements and transitions compared to baselines.

C. Robustness to Missing Modalities

Given the practical application in field study where sensor occlusion or failure is common, we evaluated model robustness under random input masking. Fig. 6 illustrates how prediction quality, measured by the Pearson correlation, changes as the proportion of masked inputs increases. Our results confirmed that GNN-MTNet maintains r of higher than 0.65 even at 30% masking, and 50% better than MM-LSTM at 40% masking. Graph structure enables information propagation from available sensors, leading to graceful degradation rather than sharp performance drops observed in baseline methods. This robustness is attributed to the GNN component, which can infer missing joint information from neighboring nodes in the anatomical graph.

D. Ablation Study

To validate the contribution of each component, we conducted an ablation study by systematically removing key modules. Table 2 summarizes the results. Removing the GNN component results in the largest performance degradation (25.8% drop in r), indicating the importance of anatomy-aware graph modeling. TAM effectively captures temporal dependencies (8.2% performance drop when TAM removed). VLA improves multi-modal aggregation by 6.9% through attending to encoder outputs per modality (r : 0.752 $\rightarrow$ 0.700). ELA refines fusion features, 5.6% contribution to overall performance (r : 0.752 $\rightarrow$ 0.710).

The ablation study shows that all components contribute to the model’s performance, with the graph-based anatomical reasoning having the largest effect. In addition, the simplified architecture with lightweight autoregressive feedback achieves higher accuracy than complex two-stage approaches, while also improving training efficiency.

Table 1: Task-specific performance comparison between our model and baselines.

Task	Model		hip fle r	hip add r	hip rot r	hip fle l	hip add l	hip rot l	arm fle r	arm add r	arm fle l	arm add l	knee ang r	knee ang l	lumbar ext	lumbar rot	ankle ang r	ankle ang l	average
Task 1	OpenCap	RMSE ↓	29.80	2.81	10.74	29.96	3.58	9.54	35.38	7.58	34.63	7.79	22.80	22.58	27.28	3.40	12.73	11.62	17.01
		r ↑	0.40	0.66	0.33	0.41	0.50	0.20	0.42	0.38	0.40	0.41	0.51	0.51	0.38	0.17	0.26	0.26	0.39
	MM-LSTM	RMSE ↓	11.41	5.34	7.22	10.33	5.33	7.98	13.49	5.69	17.89	6.59	10.94	8.63	11.09	5.07	10.79	6.42	9.01
		r ↑	0.93	0.19	0.14	0.94	0.44	0.19	0.91	0.76	0.85	0.56	0.91	0.92	0.89	0.23	0.78	0.80	0.66
	TFN-Biomech	RMSE ↓	10.02	4.15	8.53	10.80	4.45	6.39	9.34	6.14	10.23	5.91	6.43	7.29	12.10	5.91	11.15	4.77	7.73
		r ↑	0.97	0.98	0.14	0.15	0.95	0.15	0.95	0.57	0.95	0.27	0.97	0.97	0.90	0.40	0.91	0.90	0.68
Task 2	OpenCap	RMSE ↓	11.24	3.21	6.73	11.07	4.38	9.07	8.73	5.39	9.99	5.39	6.19	6.64	11.93	4.60	8.14	6.13	7.43
		r ↑	0.97	0.97	0.52	0.19	0.96	0.19	0.96	0.65	0.96	0.44	0.96	0.96	0.86	0.18	0.88	0.91	0.72
	MM-LSTM	RMSE ↓	16.21	7.59	12.60	15.21	7.05	10.61	25.02	13.28	23.42	13.69	14.58	12.39	17.16	7.54	11.82	10.62	13.68
		r ↑	0.76	0.67	0.32	0.69	0.70	0.19	0.66	0.29	0.65	0.43	0.84	0.80	0.36	0.46	0.58	0.57	0.56
	TFN-Biomech	RMSE ↓	24.78	8.29	13.37	13.21	9.70	8.73	19.61	6.55	22.59	6.52	30.05	16.32	10.31	11.73	17.87	12.24	14.49
		r ↑	0.65	0.58	0.36	0.50	0.60	0.25	0.87	0.69	0.80	0.77	0.70	0.46	0.45	0.51	0.63	0.43	0.58
Task 3	OpenCap	RMSE ↓	9.62	4.93	8.16	9.34	4.21	7.53	8.65	3.59	8.69	6.28	10.14	8.93	9.21	7.77	10.39	8.49	7.87
		r ↑	0.89	0.85	0.72	0.66	0.88	0.57	0.94	0.78	0.93	0.74	0.92	0.88	0.31	0.76	0.85	0.79	0.78
	MM-LSTM	RMSE ↓	10.28	4.71	9.30	8.69	5.84	5.02	9.04	5.14	9.60	6.16	10.18	8.17	9.91	9.63	12.86	6.68	8.20
		r ↑	0.88	0.87	0.67	0.64	0.87	0.65	0.94	0.78	0.94	0.79	0.92	0.87	0.52	0.78	0.89	0.84	0.80
	TFN-Biomech	RMSE ↓	14.90	4.89	14.49	16.39	4.52	10.29	13.92	5.56	12.46	5.47	7.18	8.77	22.15	7.47	11.17	9.99	10.61
		r ↑	0.81	0.64	0.27	0.66	0.69	0.01	0.94	0.81	0.93	0.74	0.93	0.81	0.21	0.11	0.40	0.47	0.59
Task 3	OpenCap	RMSE ↓	24.90	6.38	10.81	12.25	8.24	7.47	19.03	6.59	20.75	6.67	26.83	9.59	11.93	8.71	15.96	8.66	12.80
		r ↑	0.54	0.63	0.42	0.27	0.45	0.09	0.78	0.73	0.72	0.73	0.65	0.54	0.35	0.29	0.58	0.34	0.51
	TFN-Biomech	RMSE ↓	9.77	4.21	9.45	10.32	5.28	4.11	9.11	4.65	9.36	6.13	7.44	7.33	11.91	6.77	12.17	5.37	7.71
		r ↑	0.85	0.86	0.51	0.19	0.77	0.37	0.93	0.71	0.93	0.73	0.88	0.81	0.42	0.48	0.74	0.69	0.68
	GNN-MTNet (ours)	RMSE ↓	9.20	3.75	7.21	10.18	3.86	6.93	8.57	4.23	9.07	6.23	6.32	7.04	11.24	5.96	10.02	7.36	7.32
		r ↑	0.89	0.77	0.58	0.24	0.80	0.21	0.93	0.61	0.91	0.67	0.92	0.88	0.47	0.44	0.72	0.60	0.67

Table 2: Ablation Study Results

Configuration	r	RMSE	rRMSE
No VLA (encoder)	0.700	9.21°	0.271
No TAM	0.690	9.26°	0.292
No GNN	0.558	11.5°	0.367
No ELA	0.710	8.30°	0.288
Full (Ours)	0.752	7.55°	0.245

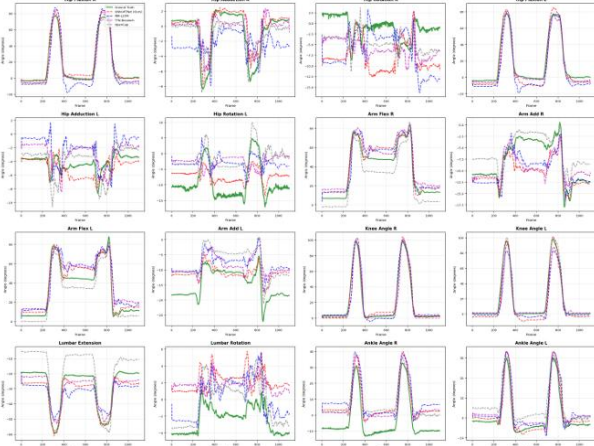
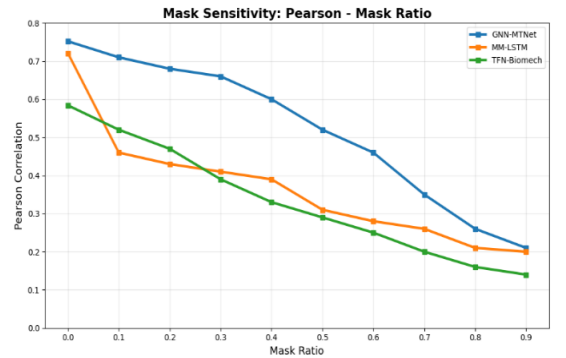


Fig. 5: Temporal prediction visualization of the representative subjects (Task 1). Green solid lines: ground truth; Red dashed: GNN-MTNet (Ours); Blue dashed: MM-LSTM; Purple dashed: TFN-Biomech; Gray dashed: OpenCap.

IV. DISCUSSION

This study proposes a novel graph-based multi-modal deep learning framework (GNN-MTNet) for joint-level kinematic estimation across three functional tasks. The model employs 70 bidirectional encoders (35 Bi-LSTM + 35 Bi-GRU) to process fine-grained temporal features from 11 IMU sensors and 13 video keypoints, combined with a 40-node graph neural network (11 IMU + 13 pose + 16 target joint nodes) using GATv2 to

capture anatomical relationships. Multi-level attention mechanisms (TAM, VLA, ELA) enable adaptive feature weighting across modalities, while a lightweight single-step autoregressive decoder maintains temporal coherence without exposure bias. The proposed model was systematically evaluated against established baseline approaches, including OpenCap, MM-LSTM, and TFN-Biomech, using joint-specific RMSE and R correlation coefficients as performance metrics. Results demonstrate that the proposed GNN-MTNet consistently outperforms baseline models in terms of average RMSE and temporal agreement, with particularly strong improvements observed for biomechanically challenging joints such as the lumbar spine, ankles, and upper limbs.

Fig. 6: Robustness analysis under random input masking. GNN-MTNet maintains higher r correlation even with 50% input channels masked, demonstrating superior robustness to missing data compared to baseline methods.

The joint-level error analysis indicates that certain joints consistently exhibited higher RMSE across all three tasks, highlighting inherent biomechanical challenges in their estimation. In general, ankle, lumbar, and hip flexion showed larger errors compared with joints such as the knee and upper limb. Several factors may contribute to these elevated errors, e.g., the occlusion challenge. In Task 1 (symmetric lifting),

ankle errors were smaller than in the other two tasks, likely because ankle motion was clearly captured by the cameras without obstruction. In contrast, during Tasks 2 and 3, which involved rung climbing, the rungs frequently occluded the ankle joint, making accurate estimation more difficult and resulting in larger errors.

For lumbar extension, the increased error may be attributed to its relatively small range of motion across all tasks, which reduces the signal-to-noise ratio and makes subtle movements more difficult for the model to predict accurately. The larger hip flexion errors observed in this study are also consistent with previous findings, where greater RMSE was reported for hip flexion in the sagittal plane during functional tasks[12] This limitation may be further amplified by the camera configuration used in this study, which employed only two cameras positioned approximately 45° relative to the subject. In contrast, a Vicon motion capture system typically uses 12 or more cameras to provide multi-angle coverage, enabling more robust tracking of hip motion.

V. CONCLUSION

This paper presents GNN-MTNet, a graph-enhanced multimodal deep learning model for full-body joint angle prediction in offshore wind energy maintenance tasks. Our key contributions include (1) a dual-encoder architecture with 70 bidirectional RNNs processing fine-grained IMU and video features, (2) GATv2-based anatomical graph reasoning that enables robust inference under sensor occlusion, and (3) a lightweight single-step autoregressive mechanism that mitigates exposure bias while maintaining temporal coherence.

Experimental results on 10 subjects performing representative occupational tasks demonstrate significant improvements: 37.3% RMSE reduction over MM-LSTM baseline, 47.4% improvement over markerless OpenCap system, and robust performance maintaining $r > 0.65$ under 30% input masking. The simplified single-stage training strategy achieves superior accuracy compared to traditional two-stage approaches while reducing training complexity.

Future work will explore (1) extension to additional occupational scenarios with more complex tool interactions, (2) real-time deployment optimization for edge devices, and (3) integration with ergonomic risk assessment systems for proactive injury prevention in field environments.

VI. REFERENCES

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